SYSTEM BLOCK DIAGRAM

ChryslerScanner V240 | integrated microSD | 259 components | 154 connected nets

VEHICLE CONNECTION

BUS INTERFACES

CONTROL & SENSING

HOST & LOGGING

OBD connector + protection

File: 06_obd-protection.kicad_sch

Battery / GND

Vehicle power input

File: 07_vehicle-input.kicad_sch

Protected supply

CCD interface

File: 01_ccd.kicad_sch

Bus data

PCI / J1850

File: 02_pci.kicad_sch

SCI stages

SCI TX14 / TX9

File: 03_sci-14-9.kicad_sch

SCI TX6 / TX12 + RX bias

File: 04_sci-6-12.kicad_sch

SCI RX7 / RX15

File: 05_sci-rx.kicad_sch

ESP32 + RF

File: 16_mcu.kicad_sch

UART / reset

SPI / detect

I2C

I/O expander + I2C

File: 18_control.kicad_sch

AW9523B | bus enables and programming control

ADC

Voltage monitoring

File: 17_adc.kicad_sch

Enables / bias

USB + reset / boot

File: 19_usb.kicad_sch

J2 USB-C | CP2102 bridge | reset / boot

Integrated SD logger

File: 22_sd-logger.kicad_sch

J4 microSD | SD_3V3 | LOG / STOP

Status

Status LED drivers

File: 20_leds.kicad_sch

POWER GENERATION & SCI PROGRAMMING

5 V / 3.3 V supplies

File: 08_power-5v.kicad_sch

VPP boost supply

File: 09_power-vpp.kicad_sch

VBB supply

File: 10_power-vbb.kicad_sch

7.5 V supply

File: 11_power-7v5.kicad_sch

SCI TX6 VBB / VPP drivers

File: 12_program-6.kicad_sch

SCI TX9 VBB / VPP drivers

File: 13_program-9.kicad_sch

SCI TX12 VBB / VPP drivers

File: 14_program-12.kicad_sch

SCI TX14 VBB / VPP drivers

File: 15_program-14.kicad_sch

Separate VBB / VPP rails feed the SCI programming stages; enables come from AW9523B.

Solid arrows: data and user interface paths. Dashed arrows: power or control relationships.

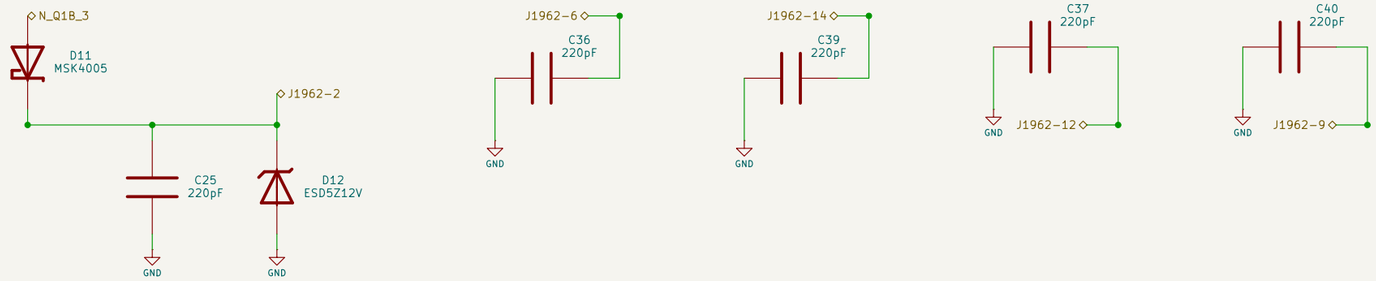
Double-click a block to open its circuit. Native hierarchical pins and named stubs carry the exact design nets.

J1 and input protection share one sheet. Solid arrows show data flow; native named stubs define electrical connectivity.

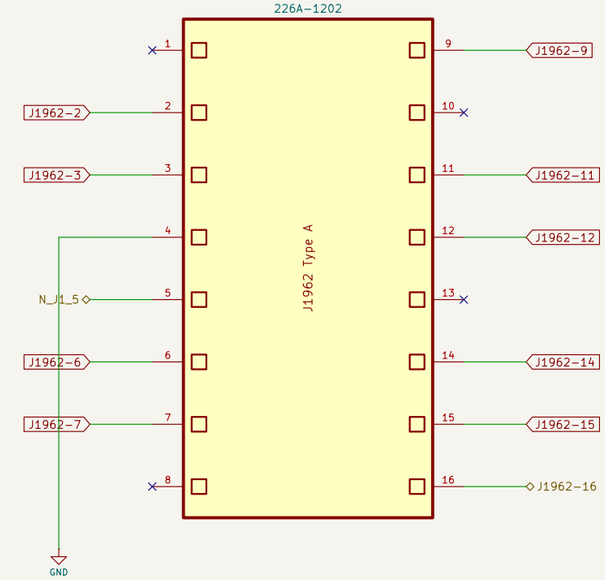
ChryslerScanner / atopile	
Sheets: /	
File: ChryslerScanner_V240_SD.kicad_sch	
Title: ChryslerScanner V240 — system block diagram	
Size: A1 | Date: 2026-10-21	Rev: V240-SD
KiCad E.D.A. 10.0.5	Id: 1/22

OBD connector and line protection

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



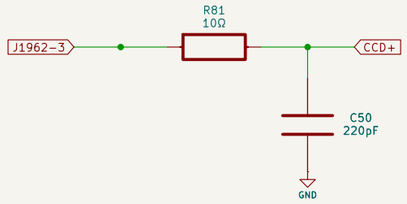
J1 — VEHICLE INPUT CONNECTOR

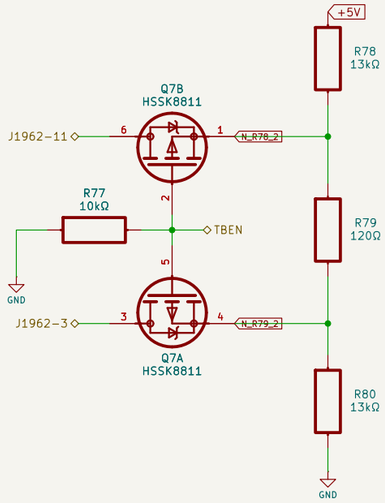


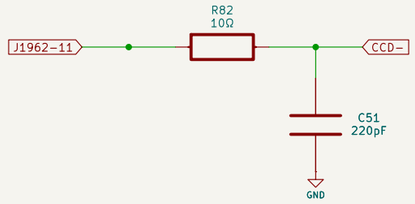
ChryslerScanner / atopile		
Sheet: /OBD connector + protection/ File: 06_obd-protection.kicad_sch		
Title: OBD connector and line protection		
Size: A3	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 2/22

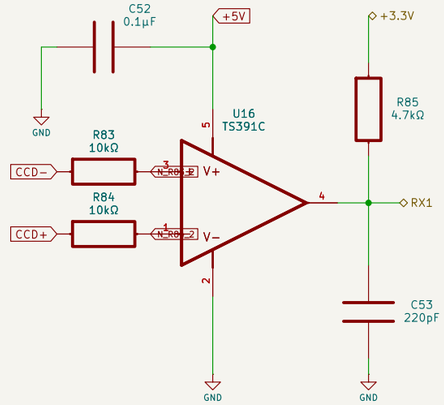
CCD interface

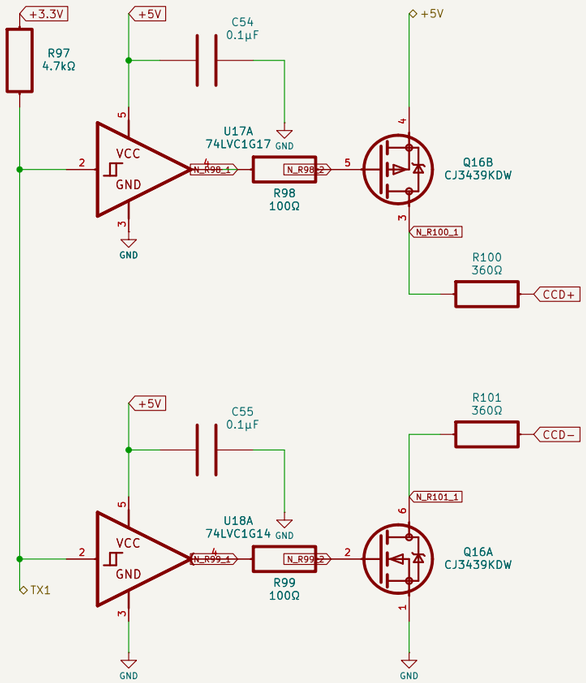
Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

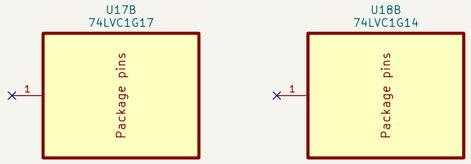












Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

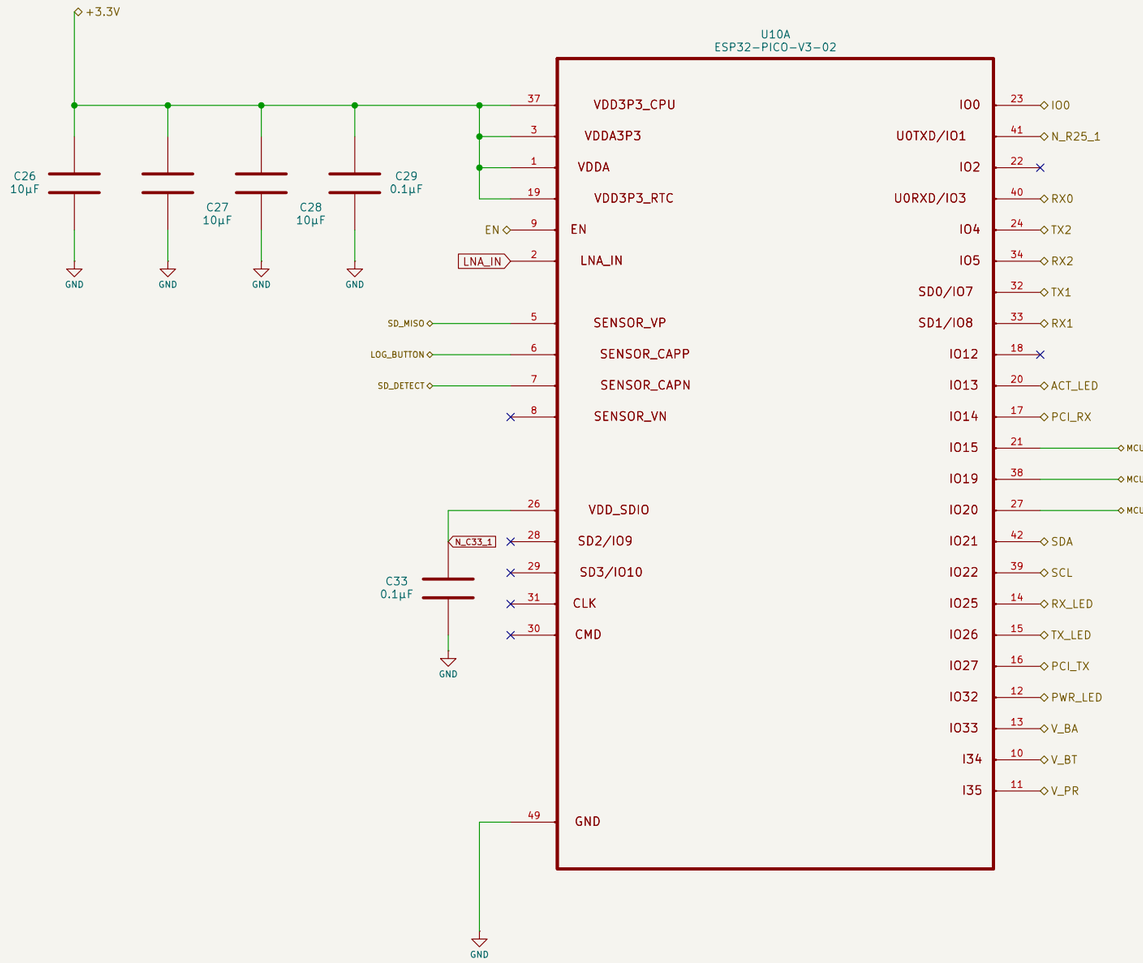
Sheet: /CCD interface/
File: 01_ccd.kicad_sch

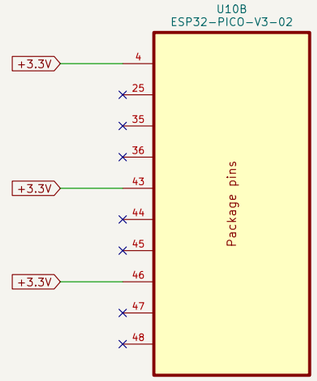
Title: CCD interface

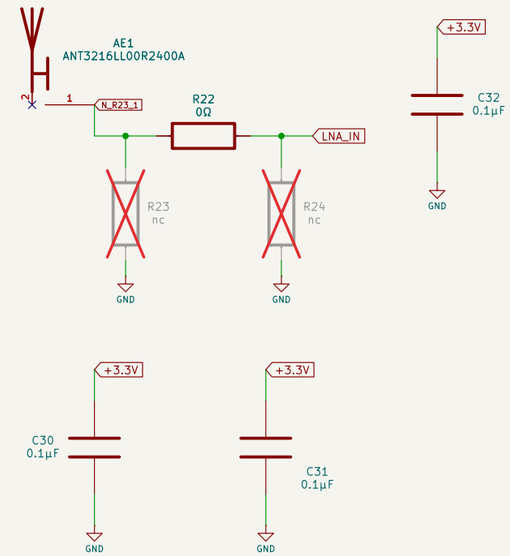
Size: A3	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 3/22

ESP32 + RF

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.







SD revision: GPIO19/20 SPI, GPIO36 MISO, GPIO15 CS, GPIO37 button, GPIO38 detect. Firmware deferred.

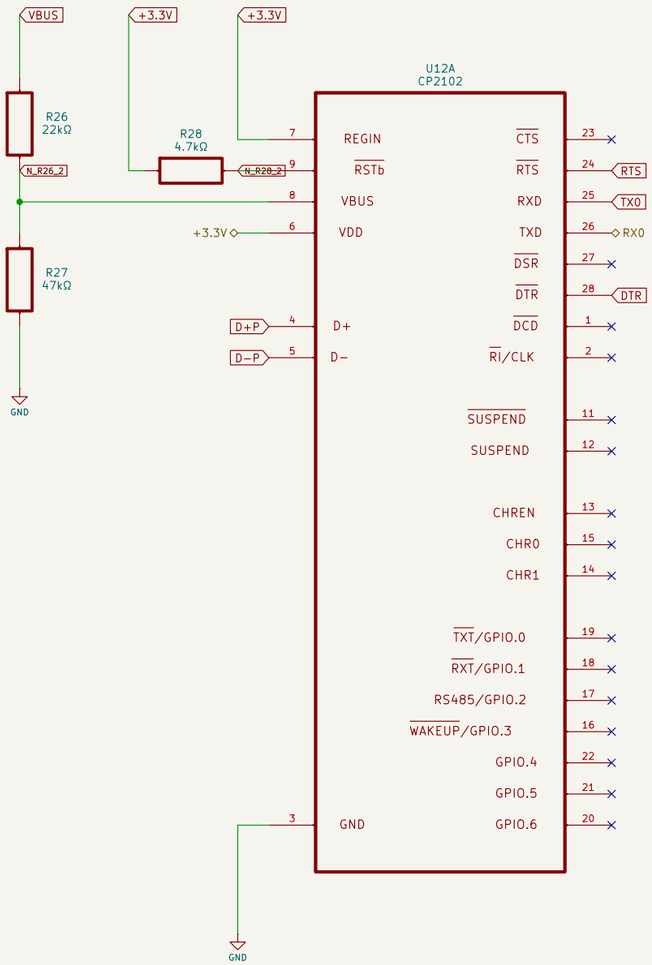
Assembly option: R23 and R24 are DNP.

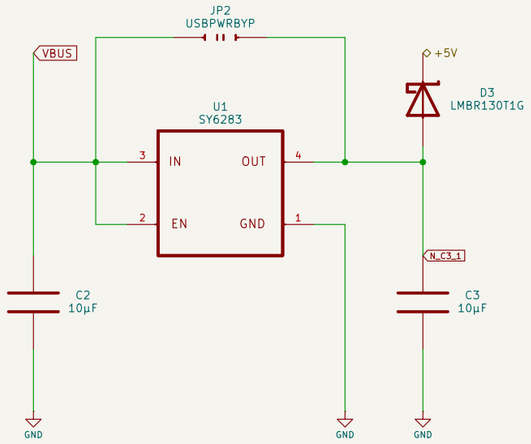
Atopile source / editable KiCad view / passive pins require electrical review.

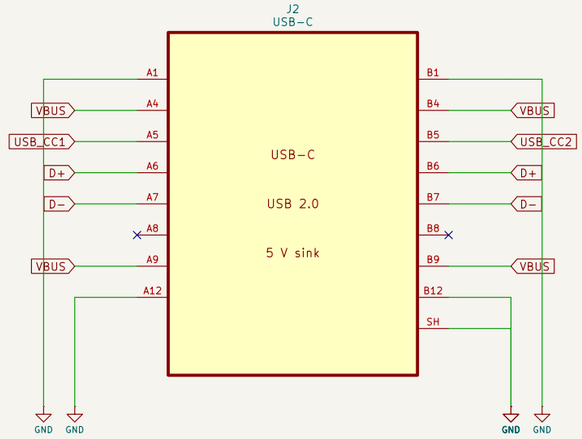
ChryslerScanner / atopile	
Sheet: /ESP32 + RF/	
File: 16_mcu.kicad_sch	
Title: ESP32 + RF	
Size: A3	Date: 2026-10-13
KiCad E.D.A. 10.0.5	**Rev: V240-C/OBD**
	Id: 4/22

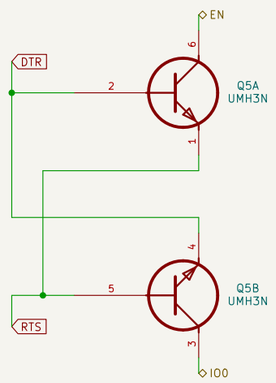
USB + reset / boot

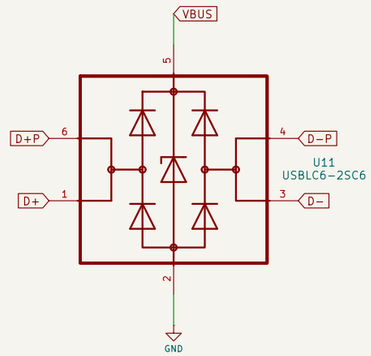
Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

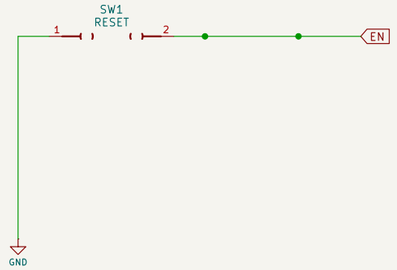


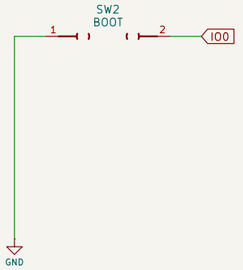


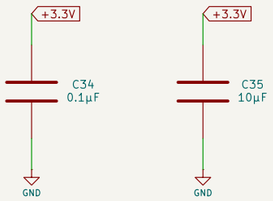


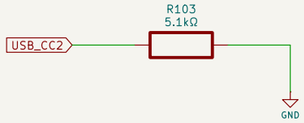




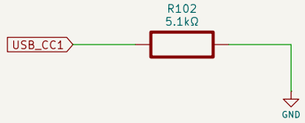


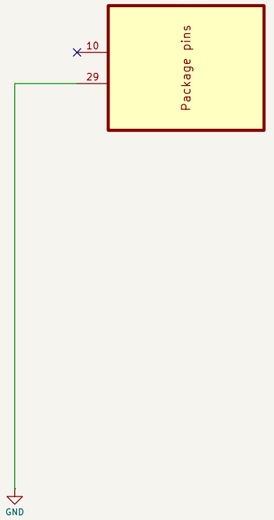












USB−C: USB 2.0 device / 5 V sink. R102 and R103: independent 5.1 kohm CC pull−downs.

Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

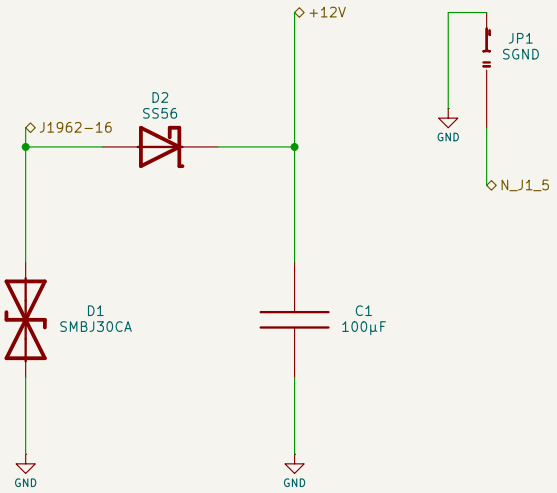
Sheet: /USB + reset / boot/
File: 19_usb.kicad_sch

Title: USB + reset / boot

Size: A3	Date: 2026−10−13	**Rev: V240−C/OBD**
KiCad E.D.A. 10.0.5		Id: 5/22

Vehicle power input

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

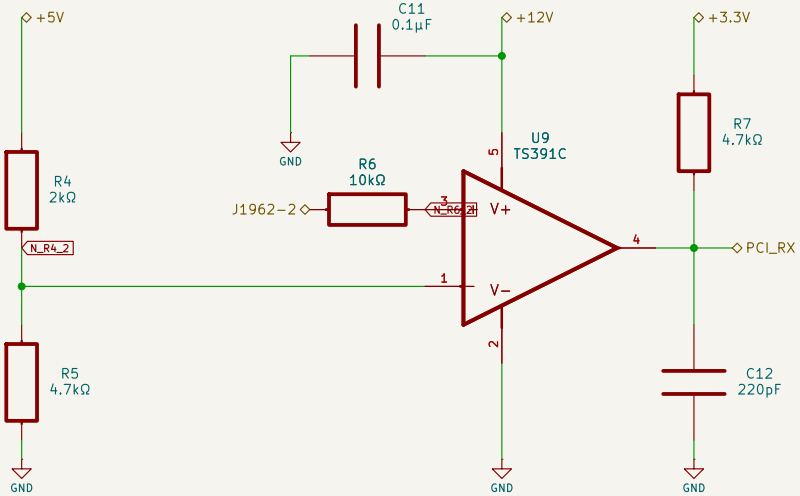
Sheet: /Vehicle power input/
File: 07_vehicle-input.kicad_sch

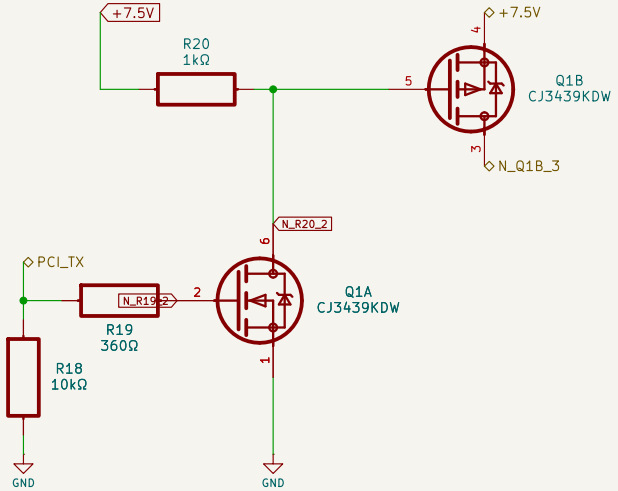
Title: Vehicle power input

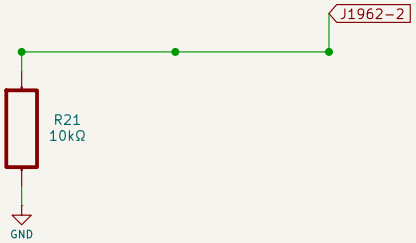
Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 6/22

PCI / J1850

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.





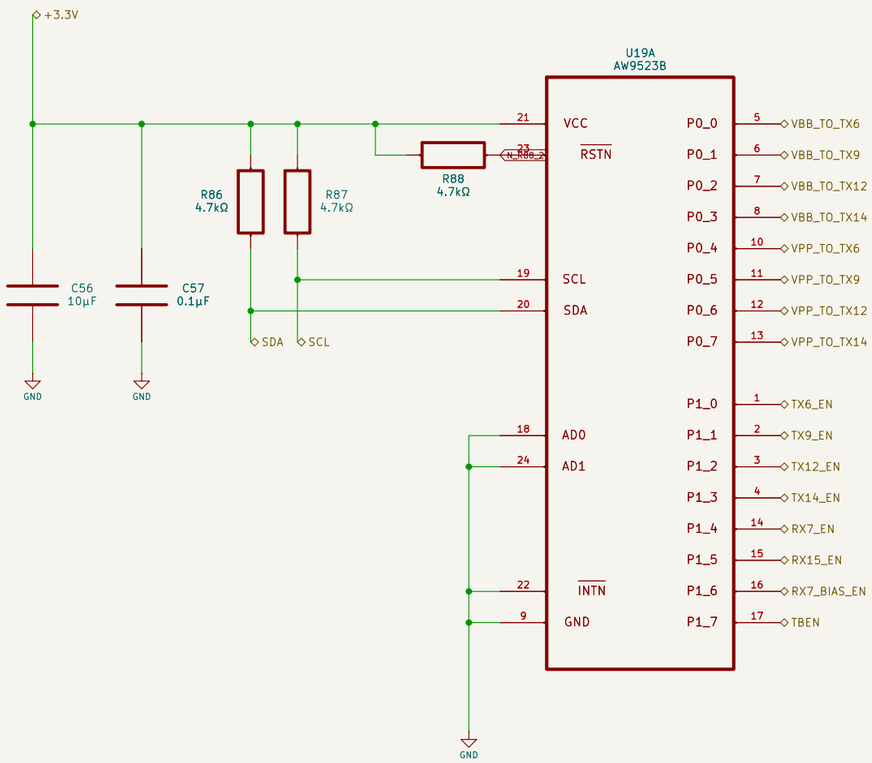


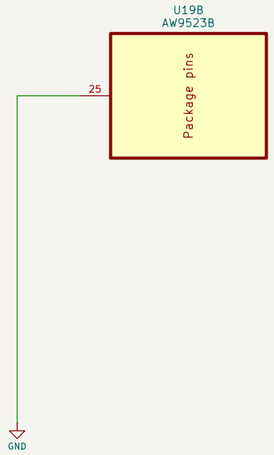
Atopile source / editable KiCad view / passive pins require electrical review.

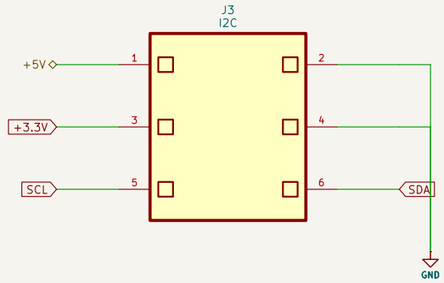
ChryslerScanner / atopile		
Sheet: /PCI / J1850/		
File: 02_pci.kicad_sch		
Title: PCI / J1850		
Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 7/22

I/O expander + I2C

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.







Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

Sheet: /I/O expander + I2C/
File: 18_control.kicad_sch

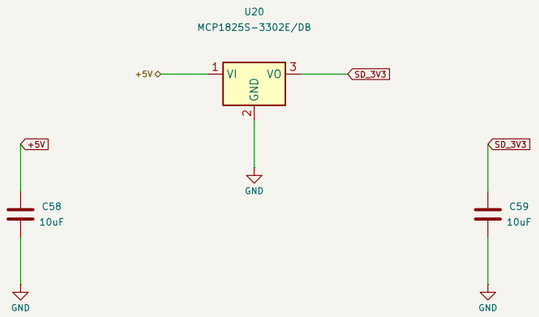
Title: I/O expander + I2C

Size: A3	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 8/22

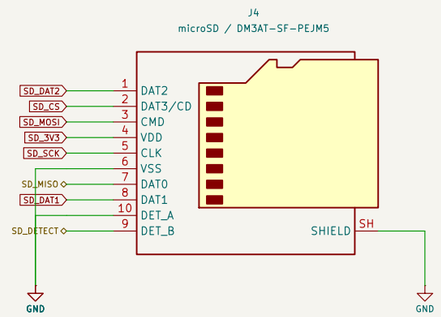
On-board microSD logger (SPI)

Existing ESP32-PICO-V3-02 | removable storage | native scanner functions retained | firmware deferred

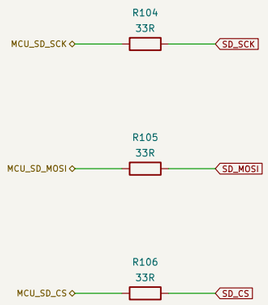
Dedicated 3.3 V SD supply from the existing +5 V rail



J4: Hirose DM3AT-SF-PEJM5 / SPI mode



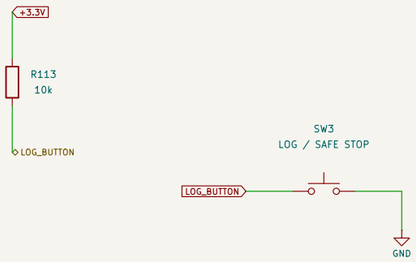
MCU-side series damping



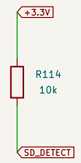
SD CMD/DAT pull-ups / MCU CS strap bias



Active-low record / safe-stop input



Card-detect input



GPIO15 CS is biased high at reset; GPIO12 and GPIO2 remain unused.

GPIO36/37/38 are input-only and have external pull-ups.

GPIO20 is package-specific; use ESP-IDF support for ESP32-PICO-V3.

SD rail transient current and total +5 V budget need hardware validation.

Bulk capacitance is not a safe-power-off hold-up supply.

Firmware deferred: this hardware revision does not yet record data.

Future status indication reuses existing ACT_LED / GPIO13.

Existing sheets use passive visualization pins; native ERC is not independent certification of the scanner circuit.

ChryslerScanner / local hardware revision

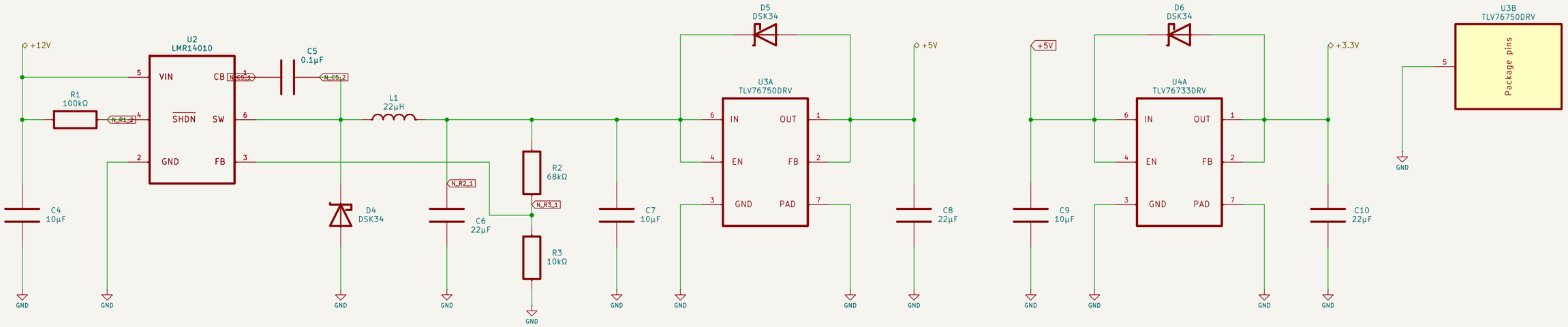
Sheet: /Integrated SD logger/
File: 22_sd-logger.kicad_sch

Title: Integrated microSD logging hardware

Size: A3	Date: 2026-10-20	**Rev: V240-SD-A**
KiCad E.D.A. 10.0.5		Id: 9/22

5 V / 3.3 V supplies

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



All TLV767 ground terminals and exposed pads connect to GND in this checked revision.

Native schematic and PCB ground connectivity are checked in the final validation reports.

Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

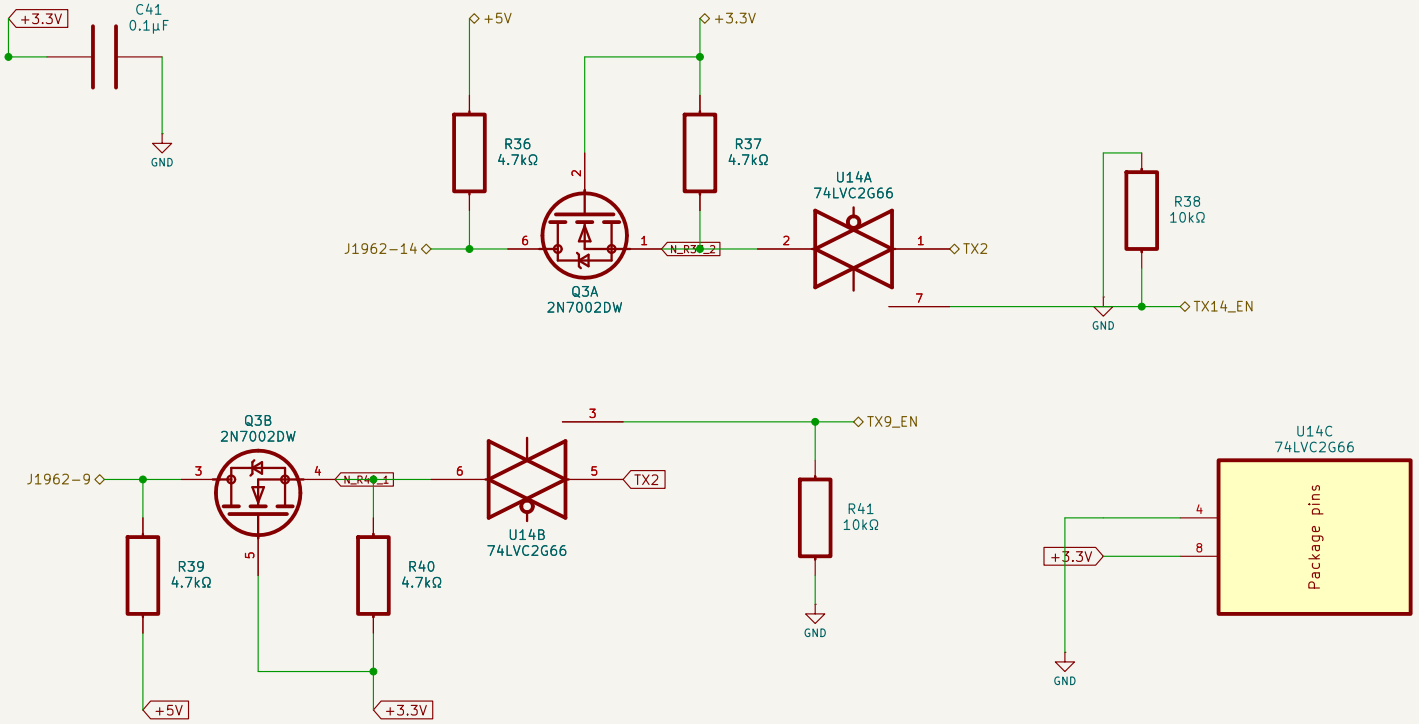
Sheet: /5 V / 3.3 V supplies/
File: 08_power-5v.kicad_sch

Title: 5 V / 3.3 V supplies

Size: A3	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 10/22

SCI TX14 / TX9

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

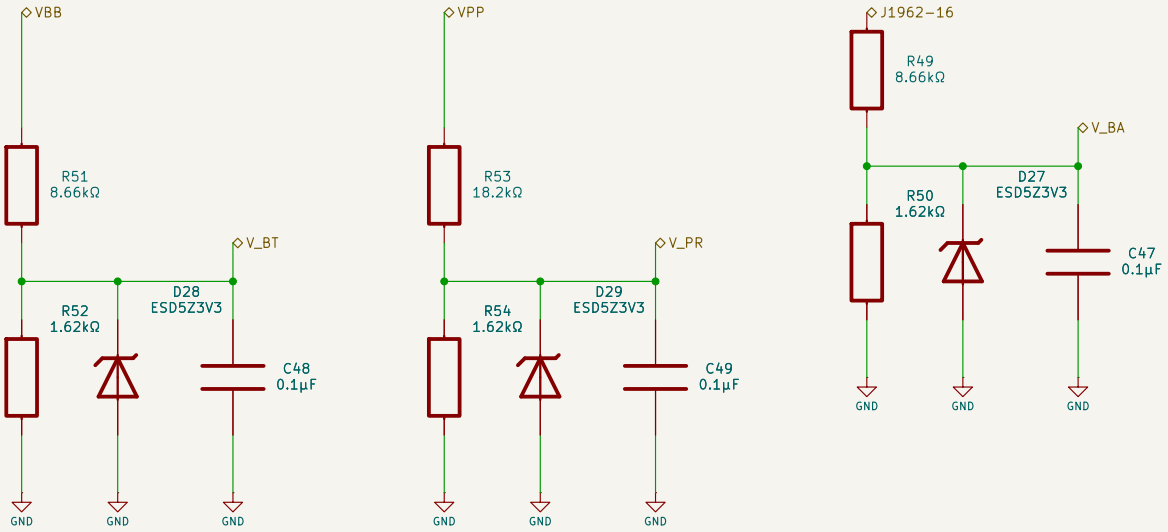


Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile		
Sheet: /SCI TX14 / TX9/		
File: 03_sci-14-9.kicad_sch		
Title: SCI TX14 / TX9		
Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 11/22

Voltage monitoring

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

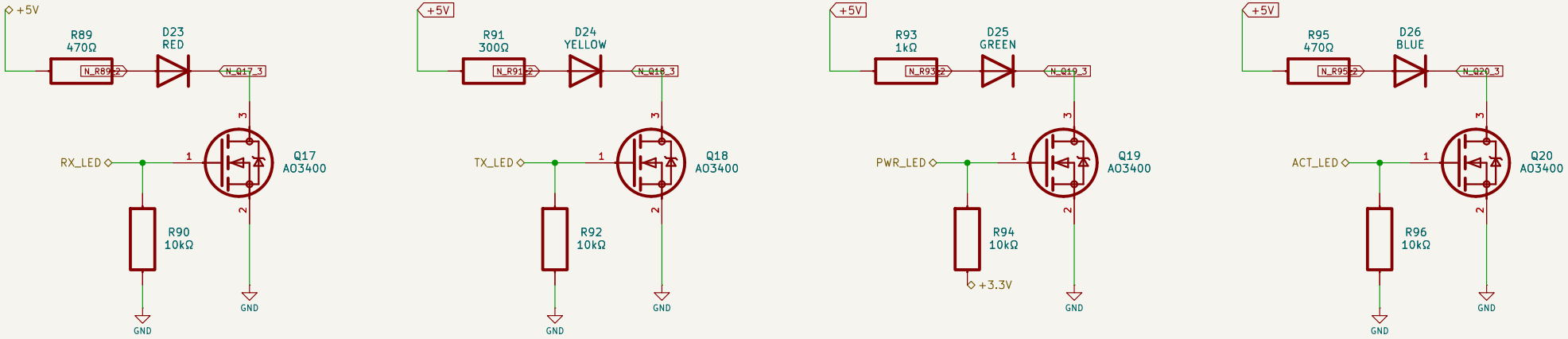


Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile		
Sheet: /Voltage monitoring/ File: 17_adc.kicad_sch		
Title: Voltage monitoring		
Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 12/22

Status LED drivers

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

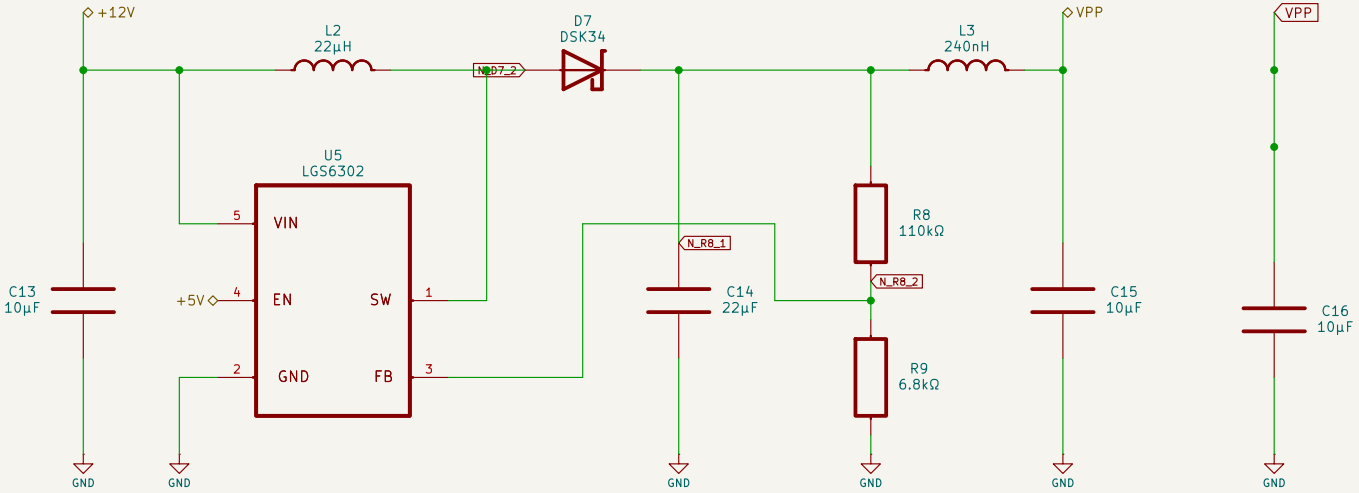
Sheet: /Status LED drivers/
File: 20_leds.kicad_sch

Title: Status LED drivers

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 13/22

VPP boost supply

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

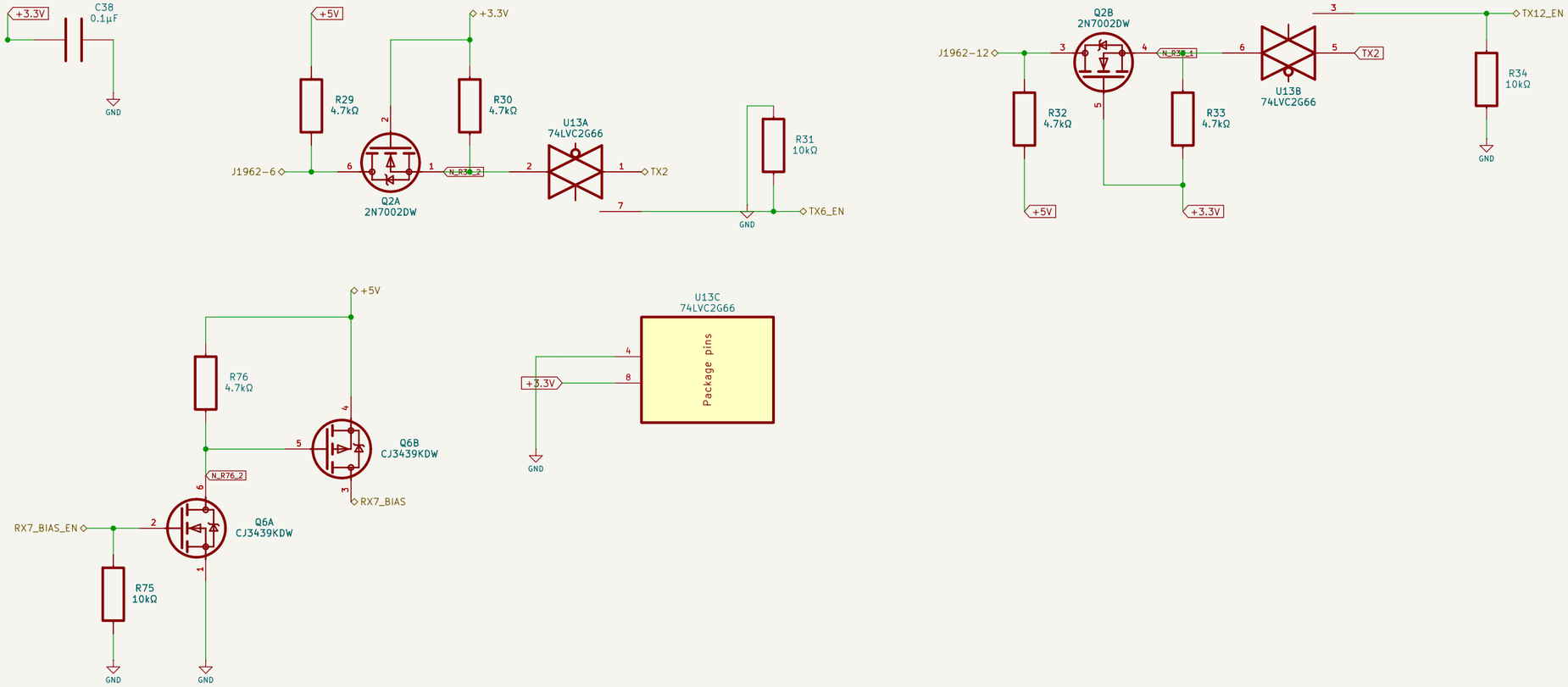


Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile		
Sheet: /VPP boost supply/ File: 09_power-vpp.kicad_sch		
Title: VPP boost supply		
Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 14/22

SCI TX6 / TX12 + RX bias

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

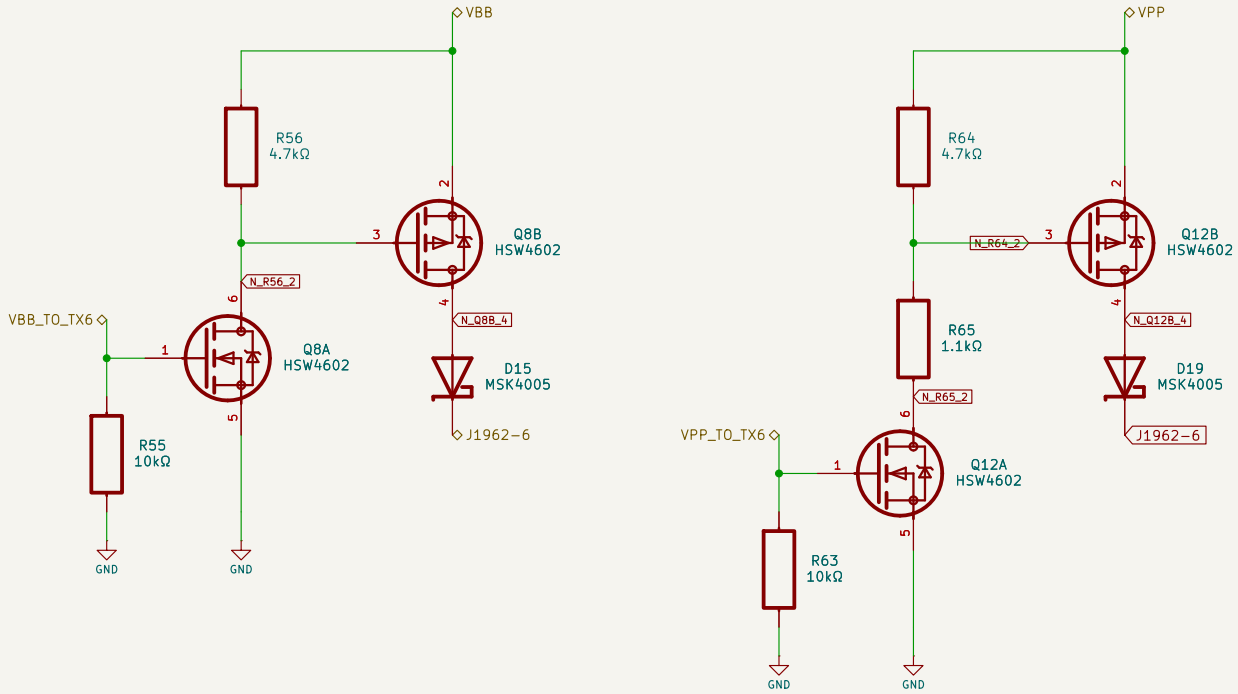
Sheet: /SCI TX6 / TX12 + RX bias/
File: 04_sci-6-12.kicad_sch

Title: SCI TX6 / TX12 + RX bias

Size: A3	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 15/22

SCI TX6 VBB / VPP drivers

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

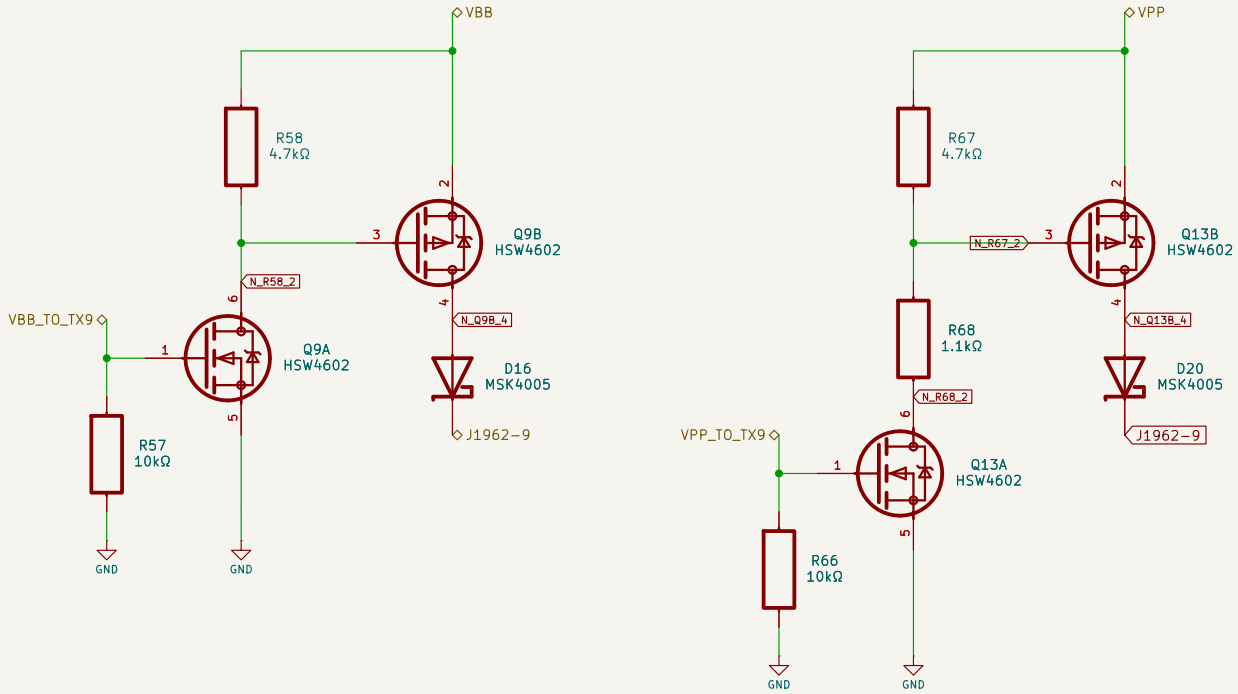


Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile		
Sheet: /SCI TX6 VBB / VPP drivers/		
File: 12_program-6.kicad_sch		
Title: SCI TX6 VBB / VPP drivers		
Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 16/22

SCI TX9 VBB / VPP drivers

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

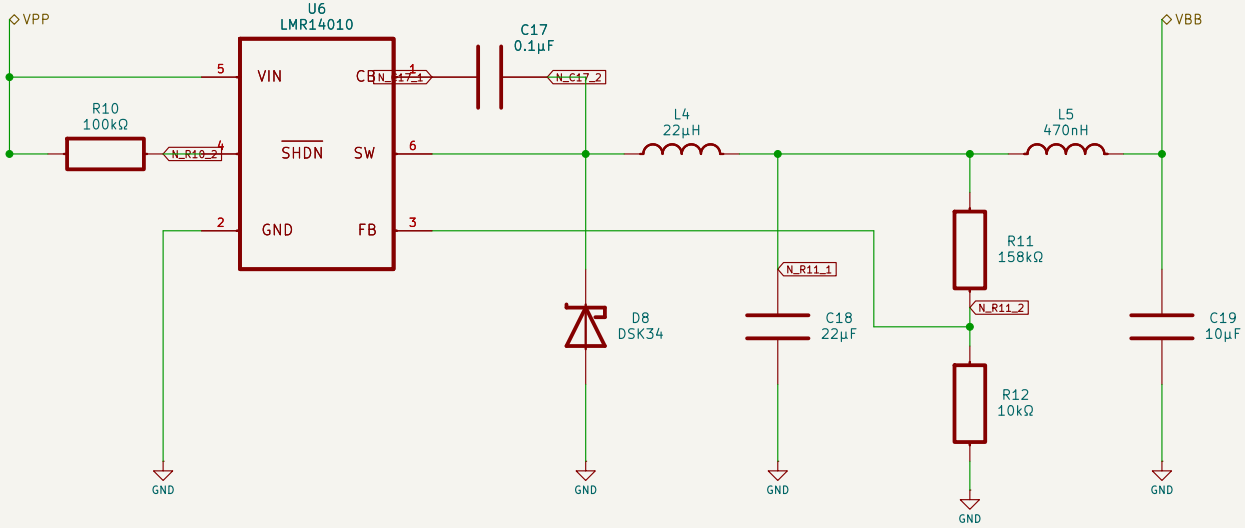
Sheet: /SCI TX9 VBB / VPP drivers/
File: 13_program-9.kicad_sch

Title: SCI TX9 VBB / VPP drivers

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 17/22

VBB supply

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

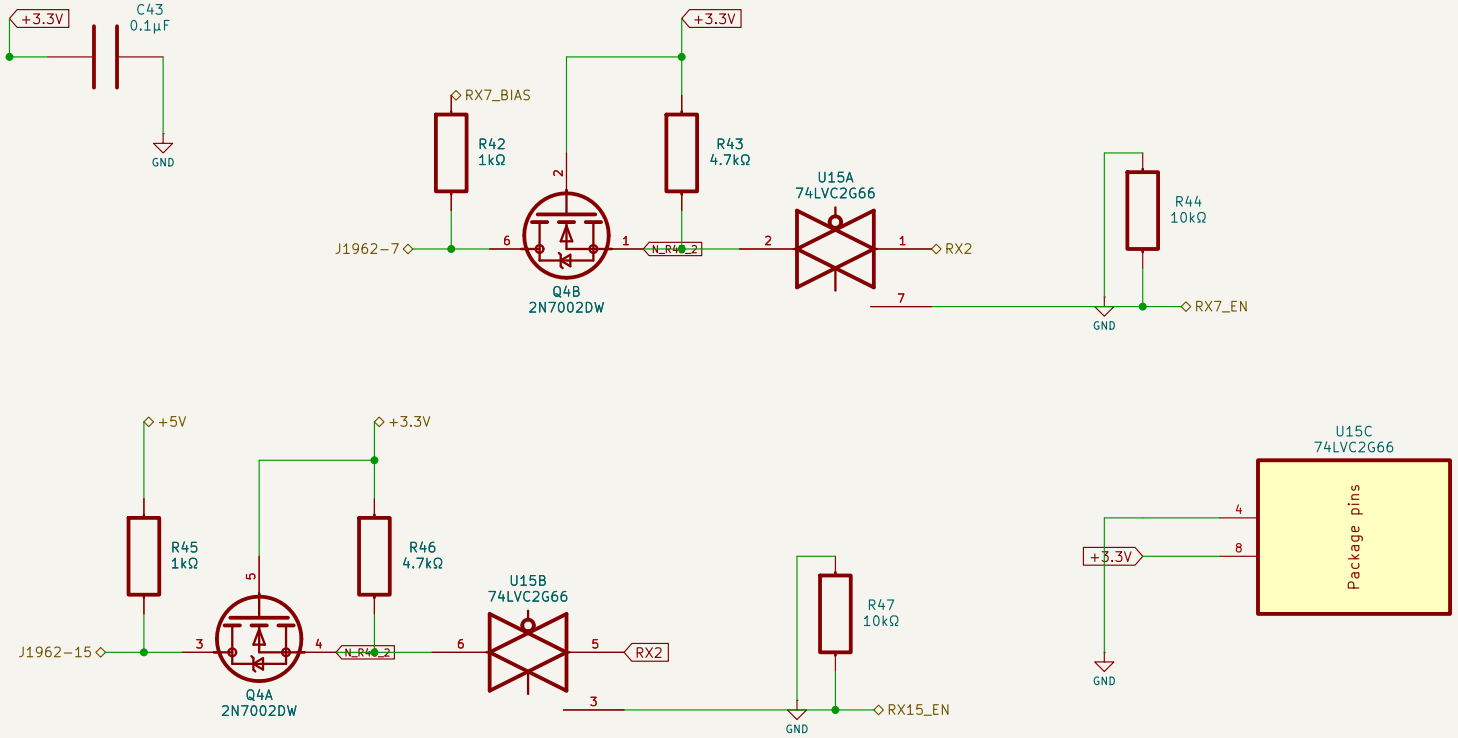


Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile		
Sheet: /VBB supply/ File: 10_power-vbb.kicad_sch		
Title: VBB supply		
Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 18/22

SCI RX7 / RX15

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

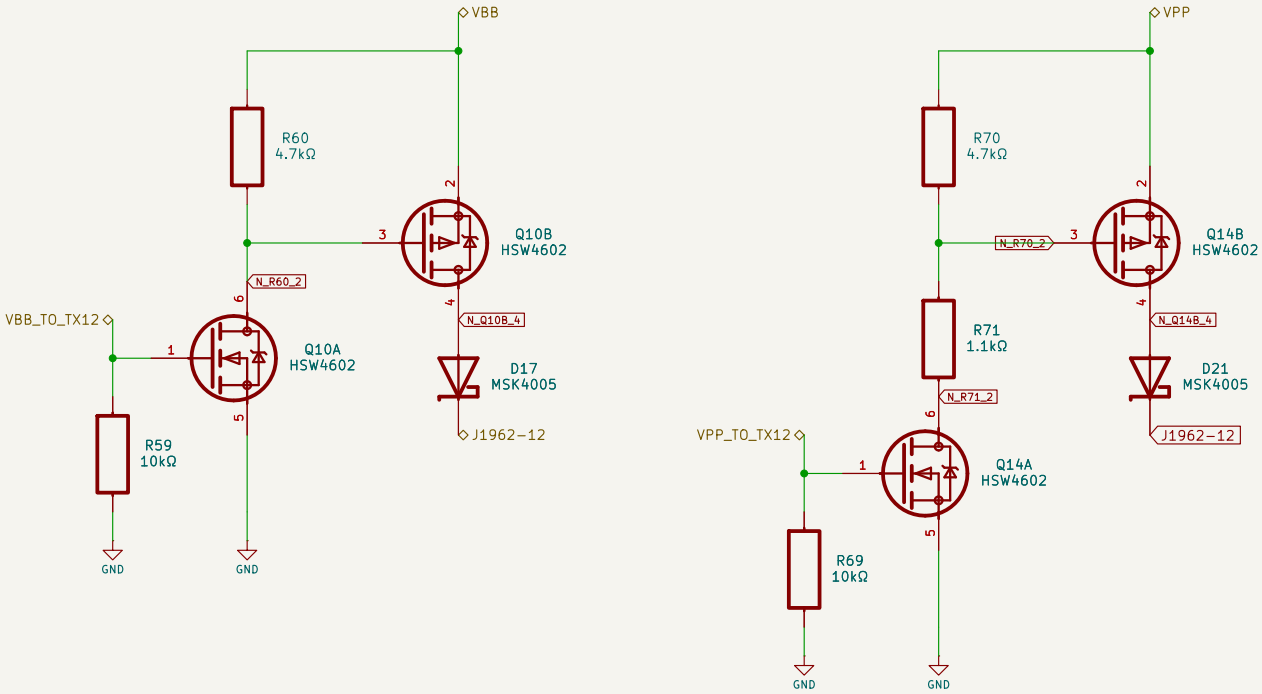


Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile		
Sheet: /SCI RX7 / RX15/		
File: 05_sci-rx.kicad_sch		
Title: SCI RX7 / RX15		
Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 19/22

SCI TX12 VBB / VPP drivers

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

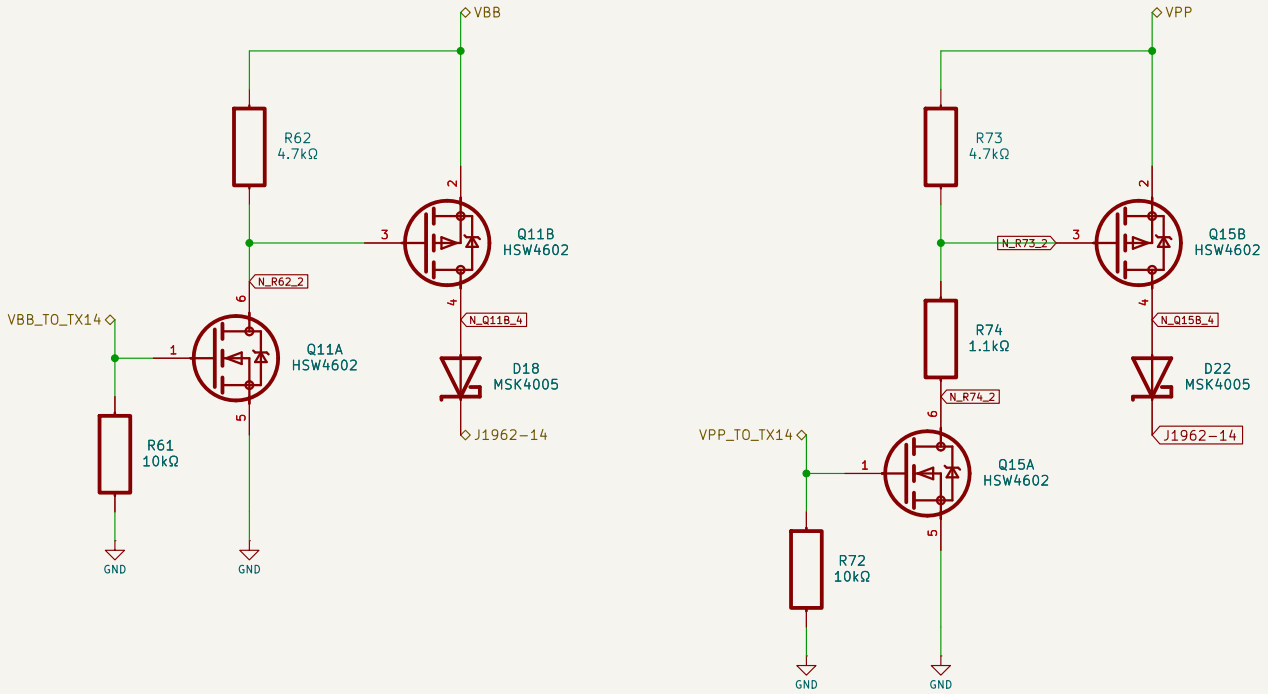
Sheet: /SCI TX12 VBB / VPP drivers/
File: 14_program-12.kicad_sch

Title: SCI TX12 VBB / VPP drivers

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 20/22

SCI TX14 VBB / VPP drivers

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

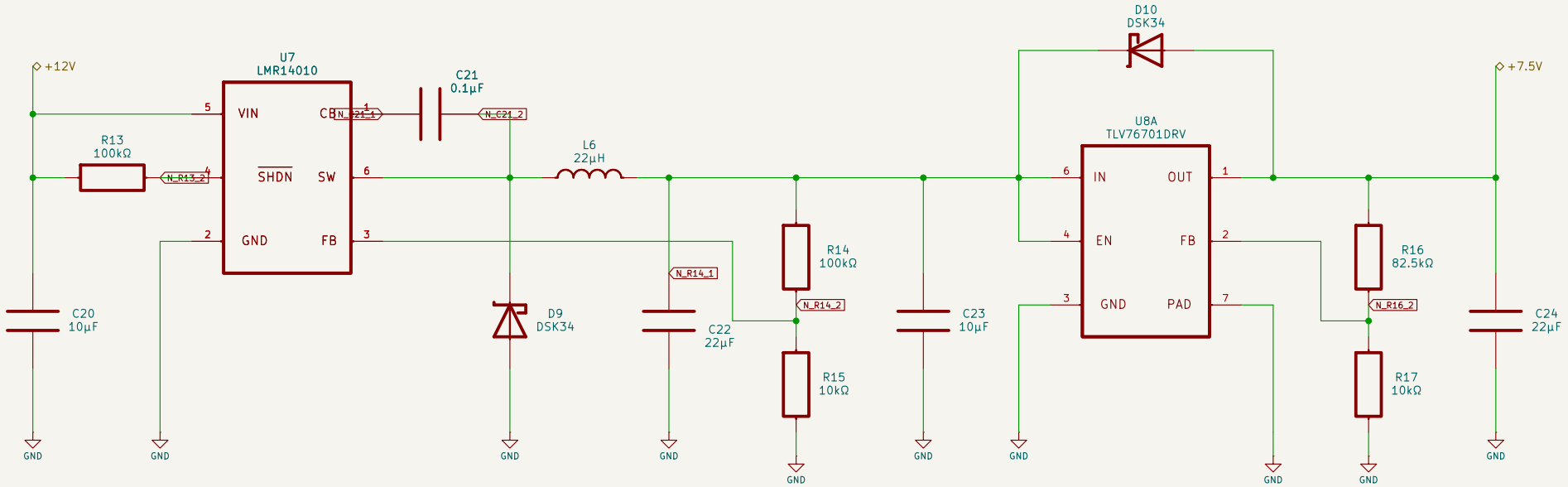
Sheet: /SCI TX14 VBB / VPP drivers/
File: 15_program-14.kicad_sch

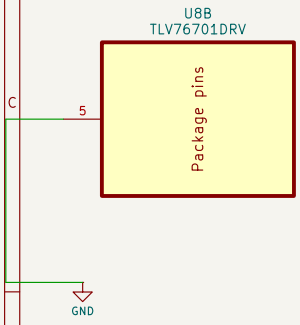
Title: SCI TX14 VBB / VPP drivers

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 21/22

7.5 V supply

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.





All TLV767 ground terminals and exposed pads connect to GND in this checked revision.

Native schematic and PCB ground connectivity are checked in the final validation reports.

Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

Sheet: /7.5 V supply/
File: 11_power-7v5.kicad_sch

Title: 7.5 V supply

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 22/22